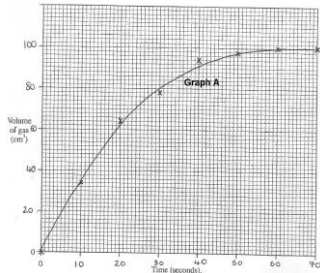
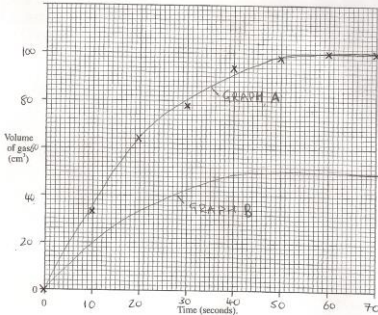


Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			 appropriate scale (1) all points plotted correctly (1) tolerance $\pm\frac{1}{2}$ square suitable curve (1) do not accept points joined with ruler [assume graph A is the graph that is plotted]		3		3	3	3
	(b)			 curve to the right of graph A (1) levelling off at 50cm³ (1)				2	2	2

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			(higher temperature means) the particles have <u>more kinetic energy / move faster</u> (1) this means there is a <u>greater frequency / chance of successful collisions</u> (1)	2			2		
	(d)			repeat the experiment / get more than one set of results / compare results with another group (1) take mean of <u>repeatable / concordant / consistent</u> results (1) accept 'reproducible' and 'reliable' results			2	2		2
	(e)			advantage – award (1) for any of following - no time delay in connecting the syringe - all of the gas produced is collected from the flask - none of the gas escapes before the syringe is connected - measuring cylinder scale is more precise than the syringe disadvantage – award (1) for any of following - measuring cylinder scale is less precise than the syringe - difficult to read the volume in the cylinder – moving water surface - some of the gas may not end up in measuring cylinder - tube may move from under the measuring cylinder NB – precision of the scale can only be credited once neutral: reference to stability or ease of setting up apparatus			2	2		2
				Question 4 total	2	3	6	11	5	9